

Dataset Fairness Evaluation Report

Stage 4: Base Fairness ML Model

Detailed Group Metrics: personal + status + sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single	140	0.7786	0.6751	0.7429	0.8214	0.0962	0.5833
female : divorced/ separated/ married	77	0.7662	0.7054	0.6753	0.7792	0.0962	0.5200
male : married/ widowed	23	0.6522	0.4889	0.6522	0.9130	0.0667	0.8750
male : divorced/ separated	10	0.7000	0.6970	0.4000	0.5000	0.2500	0.3333

Detailed Group Metrics: age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Young	93	0.7312	0.6951	0.6344	0.7097	0.1525	0.4706
Early-Middle	83	0.7831	0.5896	0.7831	0.9036	0.0615	0.7778
Late-Career	17	0.8824	0.7976	0.7647	0.8824	0.0000	0.5000
Mid-Career	57	0.7368	0.6677	0.6667	0.7895	0.1053	0.5789

Detailed Group Metrics: personal + status + sex + age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single + Young	38	0.7368	0.6955	0.6842	0.6842	0.1923	0.4167
male : single + Early-Middle	59	0.8475	0.6110	0.8475	0.9322	0.0400	0.7778

female : divorced/ separated/ married + Young	40	0.7000	0.6703	0.6000	0.7000	0.1667	0.5000
male : married/ widowed + Young	14	0.7857	0.7143	0.6429	0.8571	0.0000	0.6000
female : divorced/ separated/ married + Early-Middle	15	0.9333	0.8800	0.8000	0.8667	0.0000	0.3333
male : divorced/ separated + Young	1	1.0000	1.0000	0.0000	0.0000	0	0.0000
female : divorced/ separated/ married + Late-Career	11	0.9091	0.8070	0.8182	0.9091	0.0000	0.5000
male : single + Mid-Career	37	0.7838	0.7259	0.6486	0.8108	0.0417	0.5385
female : divorced/ separated/ married + Mid-Career	11	0.6364	0.5417	0.6364	0.8182	0.1429	0.7500
male : married/ widowed + Early-Middle	3	0.3333	0.2500	0.3333	1.0000	0.0000	1.0000
male : married/ widowed + Mid-Career	6	0.6667	0.4000	0.8333	0.8333	0.2000	1.0000
male : divorced/ separated + Early-Middle	6	0.6667	0.6667	0.3333	0.6667	0.0000	0.5000
male : divorced/ separated + Mid-Career	3	0.3333	0.2500	0.6667	0.6667	0.5000	1.0000
male : single + Late-Career	6	0.8333	0.7778	0.6667	0.8333	0.0000	0.5000

Stage 4.5: Per-Attribute Fairness ML Model

Detailed Group Metrics: personal + status + sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single	140	0.7786	0.6751	0.7429	0.8214	0.0962	0.5833
female : divorced/ separated/ married	77	0.7662	0.7054	0.6753	0.7792	0.0962	0.5200
male : married/ widowed	23	0.6522	0.4889	0.6522	0.9130	0.0667	0.8750
male : divorced/ separated	10	0.7000	0.6970	0.4000	0.5000	0.2500	0.3333

Detailed Group Metrics: age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Young	93	0.7312	0.6951	0.6344	0.7097	0.1525	0.4706
Early-Middle	83	0.7831	0.5896	0.7831	0.9036	0.0615	0.7778
Late-Career	17	0.8824	0.7976	0.7647	0.8824	0.0000	0.5000
Mid-Career	57	0.7368	0.6677	0.6667	0.7895	0.1053	0.5789

Stage 4.5: Intersectional Fairness ML Model

Detailed Group Metrics: personal + status + sex + age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single + Young	38	0.7368	0.6955	0.6842	0.6842	0.1923	0.4167
male : single + Early-Middle	59	0.8475	0.6110	0.8475	0.9322	0.0400	0.7778
female : divorced/ separated/ married + Young	40	0.7000	0.6703	0.6000	0.7000	0.1667	0.5000
male : married/ widowed + Young	14	0.7857	0.7143	0.6429	0.8571	0.0000	0.6000

female : divorced/ separated/ married + Early-Middle	15	0.9333	0.8800	0.8000	0.8667	0.0000	0.3333
male : divorce d/ separated + Young	1	1.0000	1.0000	0.0000	0.0000	0	0.0000
female : divorced/ separated/ married + Late-Career	11	0.9091	0.8070	0.8182	0.9091	0.0000	0.5000
male : single + Mid-Career	37	0.7838	0.7259	0.6486	0.8108	0.0417	0.5385
female : divorced/ separated/ married + Mid-Career	11	0.6364	0.5417	0.6364	0.8182	0.1429	0.7500
male : married/ widowed + Early-Middle	3	0.3333	0.2500	0.3333	1.0000	0.0000	1.0000
male : married/ widowed + Mid-Career	6	0.6667	0.4000	0.8333	0.8333	0.2000	1.0000
male : divorce d/ separated + Early-Middle	6	0.6667	0.6667	0.3333	0.6667	0.0000	0.5000
male : divorce d/ separated + Mid-Career	3	0.3333	0.2500	0.6667	0.6667	0.5000	1.0000
male : single + Late-Career	6	0.8333	0.7778	0.6667	0.8333	0.0000	0.5000

Stage 6: Post-Mitigation ML Model (Reweighting)

Detailed Group Metrics: personal + status + sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single	140	0.7571	0.6303	0.7429	0.8429	0.0962	0.6667
female : divorced/ separated/ married	77	0.7403	0.6375	0.6753	0.8571	0.0577	0.6800

male : married/ widowed	23	0.6522	0.4889	0.6522	0.9130	0.0667	0.8750
male : divorced/ separated	10	0.5000	0.4949	0.4000	0.7000	0.2500	0.6667

Detailed Group Metrics: age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Young	93	0.6667	0.5999	0.6344	0.7742	0.1525	0.6471
Early-Middle	83	0.7952	0.5997	0.7831	0.9157	0.0462	0.7778
Late-Career	17	0.8824	0.7976	0.7647	0.8824	0.0000	0.5000
Mid-Career	57	0.7018	0.5875	0.6667	0.8596	0.0789	0.7368

Detailed Group Metrics: personal + status + sex + age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single + Young	38	0.6316	0.5522	0.6842	0.7368	0.2308	0.6667
male : single + Early-Middle	59	0.8305	0.5948	0.8475	0.9153	0.0600	0.7778
female : divorced/ separated/ married + Young	40	0.7000	0.6581	0.6000	0.7500	0.1250	0.5625
male : married/ widowed + Young	14	0.7143	0.5758	0.6429	0.9286	0.0000	0.8000
female : divorced/ separated/ married + Early-Middle	15	0.8000	0.4444	0.8000	1.0000	0.0000	1.0000
male : divorced/ separated + Young	1	0.0000	0.0000	0.0000	1.0000	0	1.0000
female : divorced/ separated/ married + Late-Career	11	0.9091	0.8070	0.8182	0.9091	0.0000	0.5000
male : single + Mid-Career	37	0.7568	0.6813	0.6486	0.8378	0.0417	0.6154

female : divorced/ separated/ married + Mid-Career	11	0.6364	0.3889	0.6364	1.0000	0.0000	1.0000
male : married/ widowed + Early-Middle	3	0.3333	0.2500	0.3333	1.0000	0.0000	1.0000
male : married/ widowed + Mid-Career	6	0.6667	0.4000	0.8333	0.8333	0.2000	1.0000
male : divorced/ separated + Early-Middle	6	0.6667	0.6667	0.3333	0.6667	0.0000	0.5000
male : divorced/ separated + Mid-Career	3	0.3333	0.2500	0.6667	0.6667	0.5000	1.0000
male : single + Late-Career	6	0.8333	0.7778	0.6667	0.8333	0.0000	0.5000

Comparative Group Metrics: Before vs After (Reweighting)

personal + status + sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
male : single	0.6751	0.6303	0.0962	0.0962	0.5833	0.6667	0.8214	0.8429
female : divorced/ separated/ married	0.7054	0.6375	0.0962	0.0577	0.5200	0.6800	0.7792	0.8571
male : married/ widowed	0.4889	0.4889	0.0667	0.0667	0.8750	0.8750	0.9130	0.9130
male : divorced/ separated	0.6970	0.4949	0.2500	0.2500	0.3333	0.6667	0.5000	0.7000

age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Young	0.6951	0.5999	0.1525	0.1525	0.4706	0.6471	0.7097	0.7742
Early-Middle	0.5896	0.5997	0.0615	0.0462	0.7778	0.7778	0.9036	0.9157
Late-Career	0.7976	0.7976	0.0000	0.0000	0.5000	0.5000	0.8824	0.8824
Mid-Career	0.6677	0.5875	0.1053	0.0789	0.5789	0.7368	0.7895	0.8596

personal + status + sex + age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
male : single + Young	0.6955	0.5522	0.1923	0.2308	0.4167	0.6667	0.6842	0.7368
male : single + Early-Middle	0.6110	0.5948	0.0400	0.0600	0.7778	0.7778	0.9322	0.9153
female : divorced/ separated/ married + Young	0.6703	0.6581	0.1667	0.1250	0.5000	0.5625	0.7000	0.7500
male : married/ widowed + Young	0.7143	0.5758	0.0000	0.0000	0.6000	0.8000	0.8571	0.9286
female : divorced/ separated/ married + Early-Middle	0.8800	0.4444	0.0000	0.0000	0.3333	1.0000	0.8667	1.0000
male : divorced/ separated + Young	1.0000	0.0000	0	0	0.0000	1.0000	0.0000	1.0000
female : divorced/ separated/ married + Late-Career	0.8070	0.8070	0.0000	0.0000	0.5000	0.5000	0.9091	0.9091
male : single + Mid-Career	0.7259	0.6813	0.0417	0.0417	0.5385	0.6154	0.8108	0.8378
female : divorced/ separated/ married + Mid-Career	0.5417	0.3889	0.1429	0.0000	0.7500	1.0000	0.8182	1.0000
male : married/ widowed + Early-Middle	0.2500	0.2500	0.0000	0.0000	1.0000	1.0000	1.0000	1.0000
male : married/ widowed + Mid-Career	0.4000	0.4000	0.2000	0.2000	1.0000	1.0000	0.8333	0.8333
male : divorced/ separated + Early-Middle	0.6667	0.6667	0.0000	0.0000	0.5000	0.5000	0.6667	0.6667

male : divorced/ separated + Mid-Career	0.2500	0.2500	0.5000	0.5000	1.0000	1.0000	0.6667	0.6667
male : single + Late-Career	0.7778	0.7778	0.0000	0.0000	0.5000	0.5000	0.8333	0.8333

Stage 6: Post-Mitigation ML Model (SMOTE)

Detailed Group Metrics: personal + status + sex

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single	163	0.8528	0.8479	0.6074	0.5706	0.1515	0.1406
female : divorced/ separated/ married	114	0.8684	0.8667	0.4737	0.4123	0.2037	0.0667
male : divorced/ separated	26	0.8846	0.7984	0.2308	0.1154	0.5000	0.0000
male : married/ widowed	47	0.8085	0.7548	0.3404	0.1915	0.5000	0.0323

Detailed Group Metrics: age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
Mid-Career	97	0.8557	0.8511	0.4536	0.3711	0.2500	0.0566
Early-Middle	98	0.8980	0.8976	0.5714	0.4898	0.1607	0.0238
Young	117	0.8120	0.8113	0.5043	0.4359	0.2542	0.1207
Late-Career	38	0.8684	0.8661	0.4211	0.4474	0.1250	0.1364

Detailed Group Metrics: personal + status + sex + age

Group	Count	Accuracy	F1 Score	Base Rate	Selection Rate	FNR	FPR
male : single + Mid-Career	53	0.8491	0.8486	0.5094	0.4340	0.2222	0.0769

female : divorced/ separated/ married + Mid-Career	28	0.8571	0.8542	0.4643	0.3929	0.2308	0.0667
male : single + Early-Middle	48	0.9375	0.9259	0.7083	0.6875	0.0588	0.0714
male : single + Young	44	0.7955	0.7686	0.7045	0.6364	0.1935	0.2308
male : divorced/ separated + Early-Middle	6	0.8333	0.4545	0.1667	0.0000	1.0000	0.0000
male : divorced/ separated + Mid-Career	7	0.7143	0.6500	0.4286	0.1429	0.6667	0.0000
female : divorced/ separated/ married + Late-Career	9	0.8889	0.8615	0.7778	0.6667	0.1429	0.0000
male : single + Late-Career	18	0.7778	0.7750	0.3889	0.5000	0.1429	0.2727
female : divorced/ separated/ married + Young	51	0.8431	0.8322	0.3922	0.3529	0.2500	0.0968
male : married/ widowed + Young	18	0.7222	0.6990	0.4444	0.2778	0.5000	0.1000
male : divorced/ separated + Late-Career	9	1.0000	1.0000	0.2222	0.2222	0.0000	0.0000
male : married/ widowed + Early-Middle	18	0.7778	0.7231	0.3889	0.1667	0.5714	0.0000
female : divorced/ separated/ married + Early-Middle	26	0.9231	0.9231	0.5385	0.4615	0.1429	0.0000
male : married/ widowed + Mid-Career	9	1.0000	1.0000	0.1111	0.1111	0.0000	0.0000
male : divorced/ separated + Young	4	1.0000	1.0000	0.0000	0.0000	0	0.0000
male : married/ widowed + Late-Career	2	1.0000	1.0000	0.0000	0.0000	0	0.0000

Comparative Group Metrics: Before vs After (SMOTE)

personal + status + sex

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
male : single	0.6751	0.8479	0.0962	0.1515	0.5833	0.1406	0.8214	0.5706
female : divorced/ separated/ married	0.7054	0.8667	0.0962	0.2037	0.5200	0.0667	0.7792	0.4123
male : married/ widowed	0.4889	0.7548	0.0667	0.5000	0.8750	0.0323	0.9130	0.1915
male : divorced/ separated	0.6970	0.7984	0.2500	0.5000	0.3333	0.0000	0.5000	0.1154

age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
Young	0.6951	0.8113	0.1525	0.2542	0.4706	0.1207	0.7097	0.4359
Early-Middle	0.5896	0.8976	0.0615	0.1607	0.7778	0.0238	0.9036	0.4898
Late-Career	0.7976	0.8661	0.0000	0.1250	0.5000	0.1364	0.8824	0.4474
Mid-Career	0.6677	0.8511	0.1053	0.2500	0.5789	0.0566	0.7895	0.3711

personal + status + sex + age

Group	F1 (Before)	F1 (After)	FNR (Before)	FNR (After)	FPR (Before)	FPR (After)	Sel. Rate (Before)	Sel. Rate (After)
male : single + Young	0.6955	0.7686	0.1923	0.1935	0.4167	0.2308	0.6842	0.6364
male : single + Early-Middle	0.6110	0.9259	0.0400	0.0588	0.7778	0.0714	0.9322	0.6875
female : divorced/ separated/ married + Young	0.6703	0.8322	0.1667	0.2500	0.5000	0.0968	0.7000	0.3529
male : married/ widowed + Young	0.7143	0.6990	0.0000	0.5000	0.6000	0.1000	0.8571	0.2778

female : divorced/ separated/ married + Early-Middle	0.8800	0.9231	0.0000	0.1429	0.3333	0.0000	0.8667	0.4615
male : divorced/ separated + Young	1.0000	1.0000	0	0	0.0000	0.0000	0.0000	0.0000
female : divorced/ separated/ married + Late-Career	0.8070	0.8615	0.0000	0.1429	0.5000	0.0000	0.9091	0.6667
male : single + Mid-Career	0.7259	0.8486	0.0417	0.2222	0.5385	0.0769	0.8108	0.4340
female : divorced/ separated/ married + Mid-Career	0.5417	0.8542	0.1429	0.2308	0.7500	0.0667	0.8182	0.3929
male : married/ widowed + Early-Middle	0.2500	0.7231	0.0000	0.5714	1.0000	0.0000	1.0000	0.1667
male : married/ widowed + Mid-Career	0.4000	1.0000	0.2000	0.0000	1.0000	0.0000	0.8333	0.1111
male : divorced/ separated + Early-Middle	0.6667	0.4545	0.0000	1.0000	0.5000	0.0000	0.6667	0.0000
male : divorced/ separated + Mid-Career	0.2500	0.6500	0.5000	0.6667	1.0000	0.0000	0.6667	0.1429
male : single + Late-Career	0.7778	0.7750	0.0000	0.1429	0.5000	0.2727	0.8333	0.5000